

Computer Generations & History

Lesson 49 — Technology Milestones and Association Traps

JKSSB Computer Awareness

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What Does a Generation Label Mean?

- ▶ Textbooks group computer history by the dominant hardware technology.
- ▶ A generation label is a learning classification, not a precise date boundary used identically by every source.
- ▶ For exams, learn the technology association first.

High-yield distinction

Integrated circuits (ICs) are associated with the third generation; microprocessors with the fourth.

First and Second Generations

First generation

- ▶ Vacuum tubes
- ▶ Large, heat-producing systems
- ▶ Machine-language programming

Second generation

- ▶ Transistors replace vacuum tubes
- ▶ Smaller and more reliable systems
- ▶ Assembly and early high-level languages

Third and Fourth Generations

Third generation

- ▶ Integrated circuits (ICs)
- ▶ Greater reliability and reduced size
- ▶ Operating systems and multiprogramming develop

Fourth generation

- ▶ Microprocessors
- ▶ VLSI integration
- ▶ Personal computers become widespread

Fifth Generation: Use Qualified Wording

- ▶ Introductory texts often associate the fifth generation with Artificial Intelligence (AI), parallel processing, and natural-language systems.
- ▶ This is a broad textbook grouping, not a universally fixed technology boundary.
- ▶ Do not confuse an AI application with the microprocessor milestone used for the fourth generation.

Exam method

Match the wording asked in the question to the conventional association; avoid claiming that every modern computer belongs to one sharply dated era.

India Milestone: C-DAC and PARAM 8000

- ▶ **C-DAC**: Centre for Development of Advanced Computing.
- ▶ **PARAM 8000**: India's first supercomputer, developed by C-DAC and unveiled in 1991.
- ▶ It belongs to India's high-performance computing history, not to the definition of a computer generation.

C-DAC: HPC Supercomputing Systems

Association Ladder

Milestone	Association
Vacuum tubes	First generation
Transistors	Second generation
Integrated circuits	Third generation
Microprocessors	Fourth generation
AI / parallel processing (conventional grouping)	Fifth generation

Anchor the sequence: tubes, transistors, ICs, microprocessors.

Lesson 49: Recall

- ▶ Third generation = ICs; fourth generation = microprocessors.
- ▶ First = vacuum tubes; second = transistors.
- ▶ Fifth-generation descriptions commonly mention AI, but boundaries are conventional.
- ▶ C-DAC developed PARAM 8000, India's first supercomputer, unveiled in 1991.